



Google's Sword + Shield

Google protects its own. It's 'uProxy' and 'Project Shield' aim at protecting individuals. Read on to know more: <http://dgit.in/1fTHtiv>



Idiotic Names

Some characters in games really do have the most laughable names. Read on to see what we mean: <http://dgit.in/HdAVgi>

Industry Connect

Graphically speaking

At AMD, Raja Koduri's the sole man driving the company's graphics hardware and software endeavours – a rare honour. Having worked at Apple (with Steve Jobs) before re-joining AMD as VP of Visual Computing, he's tasked with strategizing AMD's GPU architecture and hardware roadmap. We talk to him about Mantle and other gaming-related stuff. Excerpts below:

Jayesh Shinde

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Q Raja, could you tell us some more about how long Mantle has been cooking inside AMD and what was the thought process behind it?

The concepts about Mantle have been in discussion for more than five years, because that's how long some of the game developers have been coming to us and complaining about the problems on the PC and the ways to solve it. In terms of it really coming together enough for us to be able to announce it, we've been only working on it really for the past year. And the timing all came out to be good because now we're in a world where majority of the games are based on game engines, and all the game engines are now getting major overhaul and upgrades in anticipation of the new consoles. I'm sure this is a good time for them to architecturally adopt something new along with all the APIs they need to support for the game engines to be compatible on consoles, tablets, multiple APIs.

Q How specific is Mantle? Can a game developer code in multiple? It's not a runtime, so can you code partially in Mantle and partially in DX?

I think it (Mantle) has enough flexibility for those kind of interesting possibilities for the developers to do, and what makes sense and what doesn't make sense remains to be seen. The other thing I'd say is that Mantle, right now, is kind of in its first phase of experiment, there are specific game engine developers that are

working on it, playing around with it and are actually trying to do a full-fledged title like Battlefield 4. While doing all this, they will be learning. They will be like 'I needed this, I needed to be able to do this, I could have done more with this, if I had compatibility with something else or interoperability with some other APIs, etc. And all of those ideas will form the roadmap for how we move Mantle forward.

I mean there's one thing about getting excited about an API and another thing about putting all the hard work necessary to achieve maximum performance. Frankly, Mantle doesn't do any magic – as I said it's a very thin, driver-level API, which means it just gets out of the way. Its main job is not to be in the way, allowing the game engine and the game to take advantage of all our hardware capabilities.

Q Do you see a fundamental shift in AMD's priorities; as in, are they turning into a predominantly gaming company?

Yeah, I mean that's a big statement. I've never seen in my past life at ATI, the amount of focus we have on gaming and game developers like we have today ever before. And that's also one segment of the market – the whole high performance side of the PC market that's still very healthy. Despite all the doom and gloom that is written around the current PC industry, which is impacting (which I call) the 'crappy side of computers' which shouldn't have existed, actually, but they exist and they're going down because they are better devices. They're much better built, better screens. So those are killing



Raja Koduri, VP, Visual Computing, AMD

that side of the PC market. But the classic side of the PC market – the reason why we all use computers, that is still going pretty healthily, and that still needs more performance. That's where the statements that 'AMD is Gaming' is coming from.

Q Can we expect a handheld gaming device from AMD in the near future?

See the thing is, you've got to think, when you ask that question 'What problem will we be solving for our users?' Companies kind of get confused about that. They think they're solving their own problems. I mean, if you think that AMD needs to get a handheld game device just to put AMD technology in their hands, are we solving a problem that the user has with handheld gaming? That's how we look at things – if there is a problem that we think we can solve – and not just to sell more hardware. That's not how you get into these businesses. Right now, I don't see that there is a problem to be solved that isn't already covered (by Android devices for mobile gaming).

Koduri once served in a range of graphics leadership roles at AMD and formerly at ATI. He is the latest example of world-class processor engineering and design talent to join AMD. **Q**